

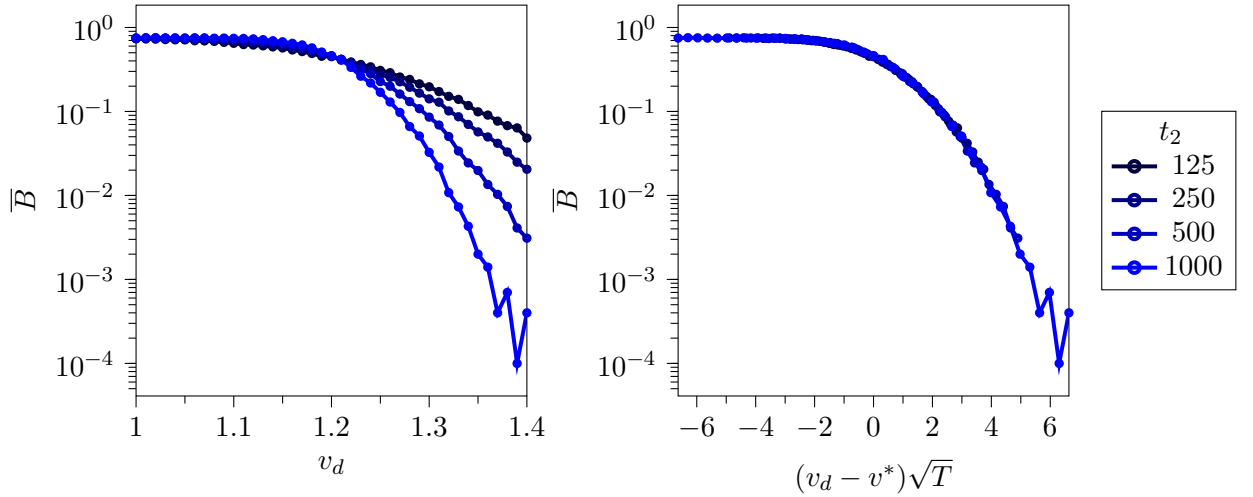


Figure 6: Left: Expanded view of the backflow phase transition near the critical point. Right: Data collapse under a finite-time rescaling. Here we set $t_1 = 100$ and $p = 0.1$. This is the result of an average over 10^4 disorder realizations.

without affecting it, the diffusive nature of this dynamics gives us the numerically observed scaling relations shown in Fig. 6. Moreover, the dynamics described by Eq. (6) are scale-free when $v = 0$. Away from this point the dynamics instead follow an error function which at long times corresponds to exponential relaxation on a timescale given by $\frac{v^2}{4D}$.

4.2 Two-dimensional medium

For the 2D problem the quantity of interest is the density of the OTOC over the entire impurity. We first extend the definition of the OTOC in Eq. 3 to have spatial dependence:

$$C^{(I)}(y, t) = \frac{1}{4} \text{tr} \left\{ \rho_0^{IM} [X_0^{(I)}(t), Y_y^{(I)}]^\dagger [X_0^{(I)}(t), Y_y^{(I)}] \right\}, \quad (8)$$

where y has been introduced to index the position on the impurity. From this we can analogously define the, now site-dependent, density: $\overline{C^{(I)}}(y, t) = \frac{2}{3} \overline{n_I(y, t)}$. The site-dependence of the operator weight on the impurity is irrelevant to the question of backflow we study here. Instead, we consider the total operator weight

$$\overline{N_I(t)} = \frac{1}{N} \sum_y \overline{n_I(y, t)}, \quad (9)$$

where N is the number of qubits within the impurity. This is the order parameter observing the phase transition in scrambling in the Markovian limit of this model [14]. The transition is characterized by the steady-state value of $\overline{N_I}$; below the critical swap rate p_c , $\overline{N_I}$ is finite and the system is in the scrambling phase, while above the critical swap rate $\overline{N_I}$ vanishes and the system is in the “absorbing phase”. This was shown to belong to the directed percolation universality class, where bond formation is promoted by the unitary gates and suppressed by the swap operations [14]. In the language of directed percolation, operator backflow can be associated with the formation of long-range temporal bonds in the temporal direction. We now consider the effect of these long-range temporal bonds on the percolation within the impurity.

Figure 7 shows the phase diagram we observe on varying the p as we take v_d across the backflow transition. There are four relevant regimes: subsonic velocities $v_d < v^*$, supersonic yet subluminal velocities $v^* < v_d \leq c$, superluminal velocities $v_d > c$, and a critical